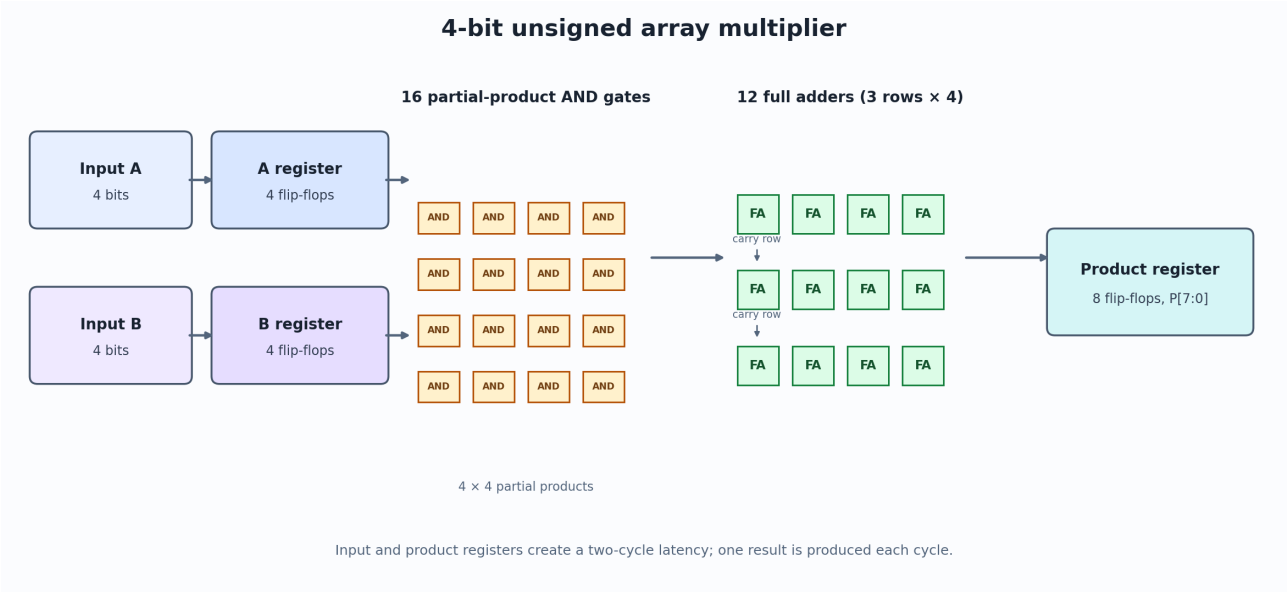
4-bit Array Multiplier

RTL design, verification, synthesis, timing analysis, and SKY130A physical design

Team member	Roll number
Dhruv Bansal	2023102048
Dataarnoor Singh Oberoi	2023102047

Design summary

The design multiplies two unsigned 4-bit inputs. Input registers capture A and B, a combinational array forms the 16 partial products and accumulates them through 12 full adders, and an 8-bit output register captures the product. The registered interface has a two-cycle latency and can accept one input pair each cycle.

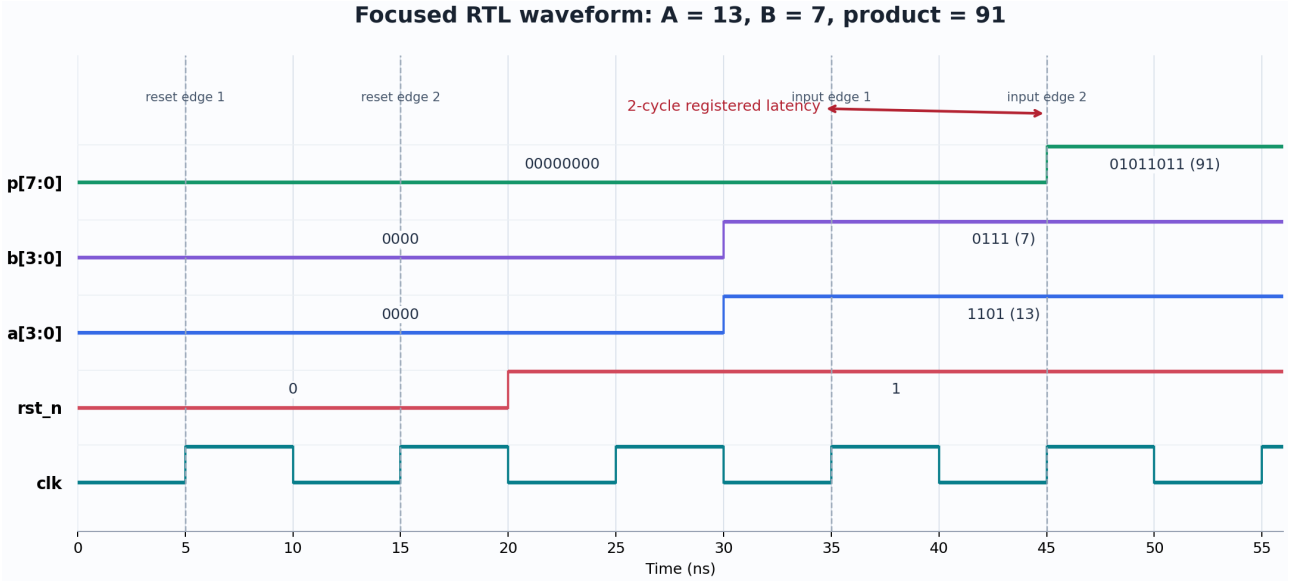


Functional verification

The testbench checks all 16 × 16 = 256 unsigned input combinations. The RTL simulation completed with PASS: 256 vectors checked, 0 errors. For the assigned hand-worked vector, A = 13 (1101 binary) and B = 7 (0111 binary), giving P = 91 (01011011 binary). A fault-injection check changed the full-adder sum expression temporarily to a ^ b; the testbench detected 95 incorrect vectors. The correct expression was restored before synthesis.

Hand-worked partial products: 00001101 + 00011010 + 00110100 + 00000000 = 01011011 (91).

RTL waveform and mapped logic



The waveform shows reset held low for two rising clock edges, then released. A=13 and B=7 are applied at 30 ns. The input register captures them at the first rising edge, and the product register presents 91 at the second rising edge.

Synthesis results

Metric	Result
Mapped standard-cell count	61
Total mapped cell area	756.976 µm²
Flip-flops	16 × sky130_fd_sc_hd__dfrtp_1
Flip-flop area	400.384 µm² (52.89%)
Most used cells	dfrtp_1: 16; nand2_1: 7; and2_0: 4 (maj3_1 also 4)
Gate-level simulation	PASS: 256 vectors checked, 0 errors

The conceptual 12 full adders are mapped into library primitives and can be combined or simplified by logic optimization; synthesized cell types therefore do not appear as twelve literal FA cells.

Static timing analysis

OpenSTA was run at the SKY130A typical corner with 1.0 ns input and output delays. The 10 ns analysis reported a worst register-to-register setup slack of 7.213 ns, giving Tmin = 10.000 – 7.213 = 2.787 ns and Fmax = 1000 / 2.787 = 358.808 MHz. A second run at the rounded test period 2.79 ns reported 0.003 ns slack on the critical register-to-register path.

Timing measurement	Value
WNS at 10 ns	7.213 ns
Estimated Tmin	2.787 ns
Fmax = 1000/Tmin	358.808 MHz
Slack at 2.79 ns test period	0.003 ns
Critical path startpoint	a_reg[1] (mapped cell _91_)
Critical path endpoint	p[6] (mapped cell _88_)
Path cell sequence	dfrtp_1 -> clkinv_1 -> o311ai_0 -> a21oi_1 -> maj3_1 -> xnor3_1 -> maj3_1 -> maj3_1 -> xor3_1 -> dfrtp_1
Total path cells / combinational cells	10 / 8 (includes launch and capture flip-flops)
Mapped full-adder functions	Approximately 5

The path has eight combinational library cells between the launching and capturing flip-flops. Several carry and sum functions are implemented by compound cells, so the logical full-adder count is not the same as the mapped serial cell count.

The report in sta/mult_array_timing.rpt contains the detailed path at the final tested period. The launch-to-capture latency is two clock periods; at Tmin, this is 5.574 ns. The steady-state throughput is one result per cycle, or about 358.808 million results per second.

Physical design results

The final fixed-die LibreLane run used a 6.5 ns clock target, an absolute 150 µm × 150 µm die, and completed after 7 timing attempts. The final metrics below are copied from the completed flow's metrics.json.

Physical-design result	Value
Clock target / run attempts	6.5 ns / 7
Die area / core area	22500.00 µm² / 17759.50 µm²
Cells after synthesis (Part B)	61
Standard-cell area / count	1224.92 µm² / 327
Added implementation cells	1550 fill; 245 tap; 3 clock buffers; 17 timing repair buffers
Routed wirelength	2361 µm
Worst setup slack (slow corner)	0.150 ns
Part C pre-route Fmax estimate	358.808 MHz
Post-route Fmax estimate	157.48 MHz
Magic / KLayout DRC errors	0 / 0
LVS errors	0

Short answers

1. Why does synthesis not report 16 AND gates and 12 full adders?

Those are conceptual RTL building blocks. The mapper decomposes the full adders into available standard cells, uses compound gates for sum/carry logic, and optimizes shared or redundant logic. Here it reports 61 mapped cells, including 16 sequential cells and 45 combinational cells.

2. What is the critical path and how does it compare with the expected 2N trend?

The worst register path starts at a_reg[1] and ends at p[6], passing through eight mapped combinational cells (including compound majority and XOR/XNOR cells). The path crosses several serial carry/sum functions, consistent with ripple-like array accumulation; the exact cell depth differs from an idealized count of about 2N full-adder delays because synthesis combines Boolean functions.

3. Why is flip-flop area a large fraction of total area?

The design uses 16 resettable flip-flops for the registered 4-bit operands and 8-bit product. Their combined 400.384 μm^2 is 52.89% of the 756.976 μm^2 mapped area, because the datapath is small while each resettable sequential cell has substantial area.

4. Why can physical-design Fmax be lower, and what cells are added?

Placement and routing add clock-tree and timing-repair buffers and parasitic interconnect, which affect setup timing. The Part C pre-route estimate is 358.808 MHz; the post-route estimate is 157.48 MHz. Placement and routing add clock-tree and timing-repair buffers and parasitic interconnect, which affect setup timing. This run also includes tap and filler cells for physical requirements.

5. What do DRC and LVS check?

Design-rule checking verifies layout geometry against foundry rules such as width, spacing, enclosure, and overlap. Layout-versus-schematic compares extracted devices and connectivity with the reference netlist. Both checks report zero errors/differences for this run.

6. How does an 8-bit array multiplier scale?

For two N-bit operands, a basic array has N^2 partial-product AND terms and approximately $N(N–1)$ full adders. At N=8 this is 64 partial-product terms and about 56 full adders, compared with 16 and 12 for N=4. Using the assignment's approximately 2N full-adder-delay estimate, path depth grows from about 8 such delays at N=4 to about 16 at N=8. Area grows approximately quadratically.

Assignment scope and evidence

This submission implements and documents a registered unsigned 4-bit array multiplier from RTL through SKY130A standard-cell physical design. The table maps the assignment's main requirements to the implementation and the evidence included with the deliverables.

Requirement	Implementation / evidence	Status
RTL architecture	16 partial products, 12 full adders, operand registers, and an 8-bit product register in rtl/.	Complete
Functional verification	10 ns testbench clock; reset asserted for two rising edges; all 256 input pairs checked; PASS result and VCD waveform.	Complete
Hand calculation and fault check	A=13, B=7 gives P=91; temporary sum fault is detected by the exhaustive testbench.	Complete
Synthesis	Yosys SKY130A mapped netlist and statistics report; 61 cells and 756.976 um^2 area.	Complete
Gate-level verification	Mapped netlist simulated with SKY130 functional cell models; 256 vectors pass.	Complete
Static timing	OpenSTA typical-corner reports at 10.00 ns and 2.79 ns, including the critical path and slack.	Complete
Physical design	LibreLane fixed 150 um x 150 um die; final metrics and GDSII; signoff DRC and LVS results.	Complete
Written analysis	Design overview, results tables, layout views, and answers to the six assignment questions in this report.	Complete

Scope and assumptions

The implemented function is unsigned multiplication of two 4-bit operands, producing an 8-bit product. The interface registers both operands and the product. The report's 8-bit discussion is an architectural scaling estimate, not a separately implemented or measured 8-bit design. Timing figures are based on the stated SKY130A corners and constraints; frequency values are derived estimates, not silicon measurements.

Final values and interpretation

The following values summarize the saved simulation, synthesis, STA, and physical-design artifacts. Derived values are shown with their calculation so the basis of comparison is clear.

Stage	Final value	Meaning
Functional test	256 / 256 pass	Exhaustive 4-bit unsigned input space.
Fault injection	95 mismatches detected	The intentionally incorrect sum logic is observable to the testbench.
Mapped synthesis	61 cells; 756.976 um^2	16 resettable flip-flops use 400.384 um^2 (52.89%).
Part C timing	10.00 - 7.213 = 2.787 ns; 1000 / 2.787 = 358.808 MHz	Tmin and pre-route Fmax estimate from typical-corner STA.
STA boundary check	2.79 ns; 0.003 ns setup slack	Near-zero positive slack at Tmin rounded to 0.01 ns.
Part D floorplan	150 x 150 um die; 17,759.5 um^2 core	Absolute die sizing in the final configuration.
Part D implementation	327 standard cells; 1,224.92 um^2; 2,361 um routed wire	Post-route standard-cell and detailed-route metrics.
Part D timing	6.5 ns target; 0.14985 ns worst slow-corner slack	Final run after seven clock-period attempts.
Post-route frequency	1000 / (6.5 - 0.14985) = 157.48 MHz	Estimate using the configured period minus worst slow-corner slack.
Signoff	Magic DRC 0; KLayout DRC 0; LVS errors 0	Saved metrics indicate clean final signoff.

Physical-design cell additions

The final metrics record 1,550 filler cells, 245 tap cells, 3 clock buffers, and 17 timing-repair buffers. These implementation cells serve layout, power, clock-tree, and timing-repair needs; they are additional to the 61 cells in the synthesized logical netlist. The final routed wirelength is taken from route__wirelength in pnr/metrics.json.

Reproduction and contribution

The source tree includes the RTL, testbench, Yosys synthesis script and outputs, OpenSTA script and timing report, LibreLane configuration, final metrics and GDSII, report source, and visual evidence. The README gives the ordered commands and expected results for each stage. The per-run working directory is intentionally excluded from the deliverables; the root-level metrics and GDSII are the final copied outputs.

Stage	Reproduction command / input	Expected artifact
RTL simulation	iverilog -g2012 -o sim tb/tb_mult.v rtl/full_adder.v rtl/mult_array.v; then vvp sim	PASS summary; mult.vcd
Synthesis	yosys -c synth/synth.tcl	synth/mult_array_netlist.v; synth/mult_array_stat.rpt
STA	CLOCK_PERIOD=10.0 sta -exit sta/sta.tcl; repeat at 2.79 ns	sta/mult_array_timing.rpt and critical-path data
Physical design	LibreLane using pnr/config.json and the selected sky130A PDK	pnr/metrics.json; pnr/mult_array.gds
Report	python3 report/build_report.py	report/report.pdf

Team contribution statement

Dhruv Bansal (2023102048) and Dataarnoor Singh Oberoi (2023102047) contributed equally to this assignment. Both members jointly contributed to the RTL design, functional verification, synthesis and timing analysis, physical-design execution, interpretation of results, and preparation of the report.

Team members

Member	Roll number	Contribution
Dhruv Bansal	2023102048	Equal contribution across the assignment.
Dataarnoor Singh Oberoi	2023102047	Equal contribution across the assignment.